PS 1

PART A

Gate 1

1) A digital image is made of many pixels and forms a visual image.
   Pixel is the smallest individual element of a digital image, each pixel corresponds to a number which has an assigned color value and these together make up the entire image.
   Image resolution tells us how many pixels there are in the form row x column
   Matrix representation of pixels is the way we align the colour numbers of each pixel in the form of a matrix to get an understanding of the pixel arrangement in the image.

2) A grayscale image only represents brightness.Each pixel has one value.in grayscale image, 0 corresponds to black and 255 to white
   RGB images store color information.
   RGB stands for red, green, blue

   (0, 0, 0) is black and (255, 255, 255) is white.

   One number tells us how much red, one how much green, and one how much blue.

   An autonomous vehicle may process them differently depending on the task. Colour can be useful for identifying coloured objects (like road markers or signs), while grayscale can be simpler for tasks such as detecting edges.

Gate 2

1) Gaussian blur is a method of smoothing an image used to remove any unwanted small details like rough texture of the road so that the more important features are more prominent and easy to detect.
2) HSV is another method of representing color apart from RGB.
   HSV= hue, saturation, value

   Hue: tells us what the color is
   Saturation:tells us how strong the colour is(A highly saturated yellow looks strongly yellow.Low saturation looks more washed out/grayish.)
   Value: tells us how bright the color is

   HSV can sometimes be more useful because it separates colour information from brightness, which can make certain colours easier to isolate.

3) An edge is basically a place in an image where there is a significant change in intensity/brightness.

In a road image, an edge might occur around:

lane markings,road boundaries,objects,buildings, etc.

Canny edge detection

Canny is an algorithm that detects edges in an image.

It takes an image and produces something where the important edges are highlighted.

4) ROI is part of the image that is relevant to the task, like the road is an important part of the image in this case but the sky is irrelevant

   this reduces false detections as Canny finds edges everywhere like sky, buildings, trees, etc, so applying an ROI highlights the important parts.

Gate 3

1) A contour is basically the outline/boundary of an object or region in an image.It helps us study the shape and location of objects or regions instead of looking at individual pixels. From a contour, we can obtain information such as the boundary, area, perimeter, shape and approximate position of the region.
2) The Hough Transform is a method for finding geometric shapes, especially lines, from detected points/edges.If many detected edge points can be associated with the same line, the Hough Transform can identify that line.

   For lane detection, we can first use an edge detector such as Canny to find possible lane edges. The Hough Transform then looks for groups of edge points that form a common straight line. These detected lines can be used as the lane boundaries.

Gate 4

1) Lane or path detection gives the vehicle information about the drivable region to maintain its position relative to the path. This information can then be used by the planning and control systems.
2) Shadows, road texture, poor lighting, missing lane markings and objects around the road can produce visual patterns that look similar to actual lane markings. It can create an edge that may be detected as a lane boundary(canny edge detection). Poor lighting can also make real lane markings difficult to distinguish. Therefore, the vision system can sometimes mistake irrelevant features for lanes, resulting in false detections.
3) Lane detection only gives the vehicle information about the lane or path. A fully autonomous vehicle needs much more information, such as the presence of obstacles, vehicles, pedestrians and other parts of the environment. It also needs

to plan a safe path and control its movement. Therefore, lane detection is an important part of perception, but it is only one part of the complete autonomous driving system.

Gate 5

1) The lane boundaries are actually roughly parallel but in the camera image, they appear to come closer together as they get farther away because of perspective.So, perspective distortion is when objects/features that are farther away appear smaller, and their positions/shapes in the image can differ from their actual physical arrangement.
2) BEV is basically a top-down representation of the road.It reduces the difficulty caused by perspective in the normal camera view and makes the position and shape of the road, lane and drivable area easier to understand. This can make navigation-related perception easier because the scene becomes more like a map.

PS 2

Part A

1.

a) A path tells you where to go but it doesn't necessarily tell you when to be at each point.A trajectory tells you where to go and how to move through that path over time.
b) A waypoint is a particular point that the vehicle is expected to pass through while following a path.

   An obstacle is something the vehicle cannot or should not travel through.

   Free space is the part of the environment where the vehicle is allowed to move.

   Start state is where the vehicle currently begins.

   Goal is where the vehicle needs to reach.

2.

a) A 2D grid map represents an environment as a collection of cells arranged in rows and columns. Each cell represents a small area of the environment and can be used to represent things such as free space or obstacles. This makes the environment easier for a path-planning algorithm to work with.
b) An occupancy grid is a type of grid map where each cell represents whether that part of the environment is free,occupied or unknown.
c) In an occupancy grid, a free cell represents an area where the vehicle can move, while an occupied cell represents an area containing an obstacle. An unknown cell represents an area whose state is not yet known because the vehicle does not have enough information about it.

3.

a) Breadth-First Search explores the environment level by level. It first checks all positions that are one step away from the starting position, then positions two steps away, and so on until it reaches the goal. When all movements have the same cost, BFS can find a shortest path in terms of the number of steps. It basically searches every where layer by layer.

b) Dijkstra’s algorithm finds a path by considering the total cost of reaching each position from the start. Unlike BFS, it can handle situations where different movements have different costs. BFS is suitable when every movement has the same cost, whereas

Dijkstra can choose a slightly longer route if that route has a lower overall cost. It goes through the cheapest options first.

c) A* is a search algorithm that combines the cost of reaching a position with an estimate of the remaining cost to the goal. This estimate is called a heuristic. The heuristic helps A* focus its search toward the goal instead of exploring equally in all directions. Therefore, A* can often find a path more efficiently than algorithms that do not use information about the goal.

4.

a) A global path describes the overall route from the starting point to the destination, usually considering the larger environment while a local motion decision deals with what the vehicle should do immediately based on its current surroundings, such as moving around a nearby obstacle. In simple terms, global planning decides the overall route, while local planning handles immediate movement.

b) An autonomous vehicle may need to re-plan when a new obstacle appears because the previously planned path may no longer be safe or possible. The vehicle has to update its map or understanding of the environment and find another path around the obstacle while still trying to reach the goal.

PS 3

Part A

Gate 1

1) A reference or setpoint is the value that the system is expected to achieve, while the measured value is the actual value obtained from the system. Feedback is the information about the actual result that is sent back to the controller. Error is the difference between the desired value and the measured value. The controller uses this error to decide what correction is needed.
2) A precomputed command assumes everything will happen as expected. But in real life, a robot may encounter uneven surfaces, disturbances or unexpected obstacles. Feedback allows it to measure what is actually happening and correct its movement accordingly.

Gate 2

1) A proportional controller produces an output that is directly proportional to the current error. It is represented by the equation $u=Kp \cdot e$

   A larger error produces a larger correction, while a smaller error produces a smaller correction.

   u = control output, such as the steering correction.

   Kp = proportional gain, which determines how strongly the controller responds.

   e = current error.

2) When Kp is increased, the controller makes stronger corrections for the error, which can make the system respond faster. However, if the gain is too high, the system may overshoot the target or oscillate.

   When Kp is decreased, the corrections become weaker, so the system may respond more slowly.

3) Steady-state error is the error that remains after the system has settled. A P controller only responds to the current error. If the remaining error is small, its correction may also be too small to overcome the forces or disturbances acting on the system so, the error may not become zero.

   A proportional controller can have steady-state error because its output depends only on the current error. As the error becomes smaller, the control output also becomes smaller.

This output may not be enough to completely correct the remaining error or overcome disturbances, so the system may settle with a small error instead of reaching the exact desired value.

Gate 3

1) A PID controller combines proportional, integral and derivative control to improve the system's response. By combining these three terms, a PID controller can correct current errors, reduce persistent errors and help prevent excessive overshooting.A PID controller combines three types of correction:

   P (Proportional): Responds to the error right now.

   I (Integral): Considers the error accumulated over time.

   D (Derivative): Considers how quickly the error is changing.

2) The proportional (P) term corrects the error that exists at the present moment. The integral (I) term considers the error accumulated over time and helps eliminate persistent errors. The derivative (D) term considers how quickly the error is changing and helps the controller respond to that change, reducing the tendency to overshoot. Together, these terms help the system follow the desired behaviour more accurately and smoothly.

Gate 4

1) Cross-track error is the sideways distance between the vehicle's current position and the desired path. It tells us how far the vehicle is away from the path.

2) Heading error is the difference between the vehicle's current heading and the desired heading along the path. It indicates whether the vehicle is facing in the correct direction to follow the path.

3) A vehicle needs to know both where it is and which direction it is facing. If it is away from the path, it needs to correct its position. If it is facing the wrong direction, it needs to

correct its orientation. Using only one of these errors may lead to poor or unstable path following.

Gate 5

1) Agent: The learner or decision-maker — the robot.

   Environment: Everything the robot interacts with.

   State: The information describing the robot's current situation.

   Action: What the robot chooses to do, such as moving left or right.

   Reward: A numerical signal indicating how good or bad the outcome is.

   Policy: The strategy the robot uses to choose an action in a given state.

   Through interaction with the environment, the agent learns which actions lead to better outcomes.

2) A Q-value tells the agent how good it expects a particular action to be in a particular state, considering the rewards it may receive in the future.

- s = the current state.
- a = the action being considered.
- Q(s,a)= the expected long-term value of taking action aaa in state sss.

For example, if a robot is near an obstacle, moving left might have a higher Q-value than moving forward because moving left may help it avoid the obstacle and eventually reach its destination safely.